Fathi's consecutive-integers bound does not extend to multitwists

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Abstract

Let S be a compact orientable surface, f a mapping class of S, and t a Dehn multitwist about disjoint curves $\alpha_1, \ldots, \alpha_n$ whose f-orbits together fill S. Ivanov asked, as Problem 5 of his 2006 list and of an earlier list dated 1994–1998, whether the integers j for which $t^j f$ is not pseudo-Anosov must lie among N_S consecutive integers, for a constant N_S depending only on S. For a single Dehn twist on a closed surface this holds with $N_S = 7$, by a theorem of Fathi (1987), which he presents as generalizing one of Long and Morton for pseudo-Anosov f; his theorem assumes only that the f-orbit of the twist curve fills. We show that this bound does not extend to multitwists: in Ivanov's orbit-filling form, with reducible base maps, no such N_S exists when S is the sphere with five boundary components, which answers Ivanov's Problem 5 negatively; his weaker question, which bounds only the number of exceptions, is untouched. For every $k \geq 1$ we give f_k and a two-component multitwist t satisfying the hypothesis for which $t^0 f_k$ and $t^{-k} f_k$ both fix an essential curve. So two exceptional exponents can lie arbitrarily far apart. The base maps f_k are themselves reducible, so this gives no counterexample with a pseudo-Anosov base map, the setting of Long and Morton. The counterexample is on the five-holed sphere only; we claim nothing about other surfaces. The proof is a short written argument; its braid-group algebra is checked in Lean, and the finite intersection and fixed-curve facts it uses are checked by two independent programs.

The proofs, the Lean formalization, the checking programs and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

Status: Written proof, with Lean and computer checks; not yet reviewed by experts.

1 Introduction

Let S be a compact orientable surface, possibly with boundary, and let $\mathrm{Mod}_S = \pi_0(\mathrm{Diff}(S))$ be its mapping class group. A mapping class is *pseudo-Anosov* if it is neither periodic nor reducible (it fixes no finite set of disjoint essential curves); see [4, 6]. Many pseudo-Anosov classes come from Dehn twists. Long and Morton [10] and then Fathi [5] studied the sequence $T_\gamma^j f$, $j \in \mathbb{Z}$, where T_γ is the Dehn twist about a curve γ. Fathi proved, for a closed surface M:

> **Theorem 0.1 of [5].** "Let f be in $\pi_0(\mathrm{Diff}(M))$ and let γ be in $\mathcal{S}(M)$. If the orbit of γ under f fills M, then $T_\gamma^n f$ is a pseudo-Anosov class except for at most 7 consecutive values of n."

Here $\mathcal{S}(M)$ is the set of isotopy classes of simple closed curves. Long and Morton [10] had proved, for pseudo-Anosov f, that $T_\gamma^j f$ is pseudo-Anosov for all but finitely many j, with no bound on the number of exceptions, as Watanabe [11, §1.1] states it; we have not read their article. Fathi's

summary says: "we generalize a theorem of Long and Morton to obtain that if f is a pseudo-Anosov diffeomorphism of an oriented surface and T_γ is the Dehn twist around the simple closed curve γ, then the isotopy class of $T_\gamma^n f$ contains a pseudo-Anosov diffeomorphism except for at most 7 consecutive values of n" [5], and Watanabe writes that Fathi "effectivised the above result of Long–Morton" [11, §1.1]. Theorem 0.1 itself assumes only that the f-orbit of γ fills, so f may be reducible; Cumplido and Wiest state it in this form, for any mapping class [2, Theorem 3]. Boyer, Gordon and Zhang later improved the constant 7 to 6 [1], as recorded by Watanabe [11, §1.1].

Ivanov asked whether Fathi's theorem extends to multitwists, with a constant depending only on the surface [8, 9]. Let t be a Dehn multitwist $T_{\alpha_1}^{\pm 1} \cdots T_{\alpha_n}^{\pm 1}$ about disjoint curves $\alpha_1, \ldots, \alpha_n$, and suppose that the curves $f^m(\alpha_i)$, $1 \le i \le n$, $m \in \mathbb{Z}$, together fill S, that is, every essential curve has nonzero geometric intersection number with one of them. His *consecutive-integers* question asks whether the j with $t^j \circ f$ not pseudo-Anosov always lie among N_S consecutive integers, for a constant N_S depending only on S; for a single twist on a closed surface this is Fathi's theorem with $N_S = 7$. His weaker *count* question asks only that at most N_S of the $t^j \circ f$ fail to be pseudo-Anosov. Both were posed as Problem 5 of his 2006 list and of an earlier list [8], for compact orientable surfaces possibly with boundary; Section 1.1 quotes the problem in full. We answer the first question, for one surface, and say nothing about the second. All curves below are essential in the usual sense (Section 2).

Theorem 1. *Let S be the sphere with five boundary components. There are essential simple closed curves a, d, c, C', C on S such that for every $k \ge 1$ and each $\varepsilon \in \{1, -1\}$ the mapping class $f_k = T_a^k T_d$ and the multitwist $t = T_a T_c^\varepsilon$ satisfy:*

1. *a and c are disjoint and not isotopic, so t is a Dehn multitwist with two components and nonzero exponents (a product of left twists when $\varepsilon = 1$);*
2. *the set $\{f_k^m(a), f_k^m(c) : m \in \mathbb{Z}\}$ fills S; already $\{a, c, f_k(a)\}$ does;*
3. *$t^0 f_k = T_a^k T_d$ fixes C' and $t^{-k} f_k = T_c^{-\varepsilon k} T_d$ fixes C, so neither is pseudo-Anosov.*

Hence the set of j with $t^j f_k$ not pseudo-Anosov contains 0 and $-k$, and no constant N_S as in the consecutive-integers question exists for this S.

Composition is right to left, as in Ivanov's $t^j \circ f$: $T_a^k T_d$ applies T_d first. The theorem holds verbatim for right-handed twists, and for $f_k t^j$ in place of $t^j f_k$, since the two are conjugate by f_k (Remark 3).

What is new, and the key verification. The mechanism is simple. Since T_a and T_c commute, $t^j f_k = T_a^{k+j} T_c^{\varepsilon j} T_d$, and the two components of t have exponents that vanish at different values of j: at $j = -k$ the twist about a disappears and at $j = 0$ the twist about c does. Each vanishing leaves a product of twists about two curves of the pentagon of Figure 1, and such a product fixes the curve of the pentagon disjoint from both. The key verification is the hypothesis: Ivanov's filling condition is on the union of the orbits of all components of t, and we must check that a, c and $f_k(a)$ already fill. This is consistent with Fathi's theorem for several curves [5, Theorem 0.2]: for curves whose f-orbits are distinct and fill a closed surface, it gives a pseudo-Anosov class once *every* exponent has absolute value at least some n, and it gives no window in which the exceptions must lie. We know of no earlier written account of this example or of an answer to the consecutive-integers question; given how short it is, a specialist may well have known it informally.

What remains open. The count question (whether at most N_S members $t^j f$ fail to be pseudo-Anosov) is untouched: our family has exceptional exponents arbitrarily far apart, and we make no claim about how many there are, nor whether each such family has only finitely many exceptions. Also untouched are the variant in which the f-orbit of each single α_i fills S (here the f_k-orbit of a stays disjoint from C'), closed surfaces, and other surfaces with boundary. Finally, the base maps $f_k = T_a^k T_d$ are reducible (they fix C'). The theorem therefore answers the question in the orbit-filling form Ivanov states, where f is arbitrary; it gives no counterexample in which f is pseudo-Anosov, the setting of Long and Morton, and says nothing about that version.

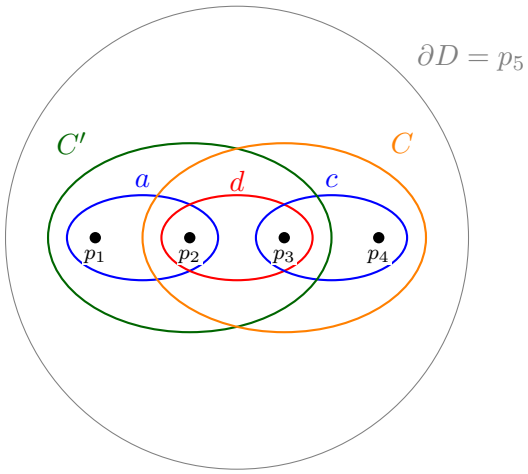


Figure 1: The five curves in the disk model: D with four punctures $p_1, \dots, p_4$ on a diameter, the boundary circle playing the role of the fifth boundary component. Each curve encloses a run of consecutive punctures: a, d, c enclose $\{1,2\}$, $\{2,3\}$, $\{3,4\}$, and C', C enclose $\{1,2,3\}$, $\{2,3,4\}$. In the cyclic order a, d, c, C', C consecutive curves meet twice and the others are disjoint. The multitwist is $t = T_a T_c^{\pm 1}$ (blue curves) and $f_k = T_a^k T_d$.

Plan. Section 1.1 quotes Ivanov's question in full. Section 2 fixes conventions and lists the standard facts used. Section 3 describes the curves and their intersection pattern, and Section 4 proves the theorem. Section 5 states exactly what the Lean proof and the two programs check, and what they do not.

1.1 Ivanov's question in full

Ivanov's list concerns "a compact orientable surface, possibly with boundary", with $\mathrm{Mod}_S = \pi_0(\mathrm{Diff}(S))$ including orientation-reversing classes [9, p. 1]. We quote Problem 5 exactly, from the arXiv version [9, pp. 4–5] (version 1, posted August 14, 2006, the only version). He writes t_α for the left Dehn twist along α.

> "**5. Dehn multitwists.** For a nontrivial circle α on S (i.e. a submanifold α of S diffeomorphic to the standard circle S^1) let us denote by t_α the (left) Dehn twist along α; $t_\alpha \in \mathrm{Mod}_S$. A Dehn multi-twist is any composition of the form $t_{\alpha_1}^{\pm 1} \circ t_{\alpha_2}^{\pm 1} \circ \cdots \circ t_{\alpha_n}^{\pm 1}$ for disjoint circles $\alpha_1, \alpha_2, \dots, \alpha_n$.
>
> **Question.** Is there a constant N_S, depending only on S, such that the following holds? Let $f \in \mathrm{Mod}_S$ and let $t = t_{\alpha_1}^{\pm 1} \circ t_{\alpha_2}^{\pm 1} \circ \cdots \circ t_{\alpha_n}^{\pm 1}$ be a Dehn multi-twist. If

> $A = \{f^m(\alpha_i) : 1 \le i \le n,\ m \in \mathbb{Z}\}$ fills S (i.e. for any nontrivial circle γ there exists an $\alpha \in A$ such that the geometric intersection number $i(\gamma, \alpha) \neq 0$), then only a finite number of elements of the form $t^j \circ f$, $j \in \mathbb{Z}$, are not pseudo-Anosov, and the corresponding j are among some N_S consecutive integers.
>
> By a theorem of A. Fathi [Fath], if t is a Dehn twist, then this is true for $N_S = 7$. This theorem of A. Fathi was preceded by a theorem of D. Long and H. Morton [LM] to the effect that for a Dehn twist t, only a finite number of element $t^j \circ f$ are not pseudo-Anosov under the above assumptions, without a uniform bound on the number of exceptions.
>
> A weaker version of this question is still interesting: under the same conditions, is it true that no more than N_S of $t^j \circ f$ are not pseudo-Anosov?"

Long and Morton's theorem, as noted above, assumes that f is pseudo-Anosov, which is stronger than the filling hypothesis in Ivanov's summary of it. Ivanov goes on to call Fathi's paper "one of the deepest and the most underappreciated works in the theory of the mapping class groups" and adds: "The above question may be considered as a test of our understanding of Fathi's ideas" [9, p. 5]. The question already appears, in essentially the same words, as Problem 5 of his earlier list [8], which also states the weaker version. That list is headed "1994–1998, 2014"; Ivanov writes there that its original version was written in early 1994 and that updates were added until 1998, marked as such, and Problem 5 carries no update, but we have not seen the 1994 original. Watanabe restated it in 2016, describing it as "a Long's and Penner's like extension on Fathi's theorem" [11, §1.1]. Zhang discussed whether the twist can be replaced by a multitwist "at a cost of replacing the number 7 by an undetermined but universal number", citing Ivanov's manuscript *Mapping class groups* of December 21, 1998, in a preprint first posted August 6, 2007 [13]. For a special class of pseudo-Anosov maps (those that become trivial when a chosen puncture is filled in) and multitwists satisfying a non-compressibility condition, it claimed that every composition is pseudo-Anosov, so that no exceptions occur; the author withdrew the preprint in January 2008. In a later paper [14], Zhang showed that for any Dehn twist t_c on a punctured surface there are pseudo-Anosov maps f_m with $f_m \circ t_c^k$ pseudo-Anosov for all k once m is large, and deduced a corollary for multitwists on two curves. Both works concern pseudo-Anosov base maps and give positive results; our example is of a different kind. So there are two questions: the *consecutive-integers* question, which asks for Fathi's conclusion with N_S in place of 7, and the weaker *count* question. We answer the first, for one surface, and say nothing about the second.

2 Conventions and standard facts

Throughout, S is the compact sphere with five boundary components. Ivanov's Mod_S is $\pi_0(\mathrm{Diff}(S))$, so isotopies may move the boundary. Isotopy classes of curves, the action of mapping classes on them, geometric intersection numbers and the Nielsen–Thurston types are therefore those of the five-punctured sphere $S_{0,5}$, and we work there. A simple closed curve is *essential* if it bounds neither a disk nor a disk with one puncture; on $S_{0,5}$ this means that each side contains at least two of the five punctures. This is Ivanov's "nontrivial" circle, which he describes in his Problem 6 as "not bounding a disk and non deformable into the boundary" [9]. (With a weaker reading the filling hypothesis could never hold, since an inessential curve meets nothing.) Ivanov's Mod_S includes orientation-reversing classes; our example uses orientation-preserving ones only.

We write $i(x, y)$ for the geometric intersection number of isotopy classes and T_x for the left Dehn twist about x. We use these standard facts:

(D) If $i(x,y)=0$ then $T_x(y)=y$ and $T_xT_y=T_yT_x$, since the twists have disjoint supports.

(I1) $i(T_x^m(y),y)=|m|\,i(x,y)^2$ for all $m\in\mathbb{Z}$ [4, Prop. 3.2].

(I2) $\big|\,i(T_x^m(y),z)-|m|\,i(x,y)\,i(x,z)\big|\le i(y,z)$ for all $m\in\mathbb{Z}$ [4, Prop. 3.4].

(N) $i(h(x),h(y))=i(x,y)$ for every mapping class h.

(P) A pseudo-Anosov class fixes no isotopy class of essential curve (Nielsen–Thurston classification [4, Thm. 13.2]; see also [6]).

(Q) A pair of pants (a sphere with three boundary components or punctures) contains no essential curve that is not parallel to a boundary component or puncture. Indeed, a simple closed curve in it separates it, and one of the two sides contains at most one of the three boundary components or punctures, so that side is a disk or an annulus.

3 The curves

Model S as the closed disk D with four punctures p_1,p_2,p_3,p_4 on a horizontal diameter, in this order; the boundary circle ∂D plays the role of the fifth puncture p_5 (collapsing it to a point gives $S_{0,5}$). For a set I of consecutive indices, the *round curve* about I is the boundary of a convex neighborhood of $\{p_i : i\in I\}$ that contains no other puncture. Our five curves (Figure 1) are

$$a=\mathrm{round}\{1,2\},\quad d=\mathrm{round}\{2,3\},\quad c=\mathrm{round}\{3,4\},$$

$$C'=\mathrm{round}\{1,2,3\},\quad C=\mathrm{round}\{2,3,4\}.$$

Each separates the five punctures into sets of sizes 2 and 3, so each is essential; and they induce five different partitions of $\{1,\dots,5\}$, so they are pairwise non-isotopic. Equivalently, with $p_1,\dots,p_5$ placed at the corners of a convex pentagon in this cyclic order, a, d, c, C', C are the boundaries of thin neighborhoods of the sides p_1p_2, p_2p_3, p_3p_4, p_4p_5, p_5p_1.

Lemma 2 (the pentagon pattern)**.** *In the cyclic order (a,d,c,C',C), consecutive curves have geometric intersection number 2 and non-consecutive curves are disjoint:*

$$i(a,c)=i(a,C')=i(d,C')=i(d,C)=i(c,C)=0,$$
$$i(a,d)=i(d,c)=i(c,C')=i(C',C)=i(C,a)=2.$$

Proof. Round curves about runs I, J are disjoint when $I\subseteq J$, $J\subseteq I$ or $I\cap J=\emptyset$: choose nested or disjoint convex neighborhoods. This gives the five zeros. In each of the other five pairs the runs overlap partially, and the round curves can be drawn meeting in two points, as in Figure 1. Each region bounded by one arc of each curve contains a puncture, so there is no bigon, and by the bigon criterion [4, Prop. 1.7] the curves are in minimal position, so the intersection number is 2. □

Section 5 describes two independent exact checks of the pattern, in the form $i(u,v)=0 \iff T_u(v)=v$ (which follows from (D) and (I1)).

4 Proof of Theorem 1

We prove the three items of the theorem in turn and then conclude. Fix $k\ge 1$ and $\varepsilon\in\{1,-1\}$, and put $f=f_k=T_a^kT_d$ and $t=T_aT_c^{\varepsilon}$.

(1) The multitwist. By Lemma 2, a and c are disjoint, and they are not isotopic. Both exponents in t are nonzero, and for $\varepsilon = 1$ the multitwist t is a product of left twists.

(2) The orbits fill. We first show that an essential curve γ with $i(\gamma, a) = i(\gamma, c) = 0$ is isotopic to a or to c. Suppose it is isotopic to neither. Cutting S along the disjoint curves a and c leaves three pairs of pants: one bounded by a, p_1, p_2; one by c, p_3, p_4; and one by a, c, p_5. Realize a, c, γ by geodesics of a complete hyperbolic metric; distinct simple closed geodesics are in minimal position [4, §1.2], so γ is disjoint from $a \cup c$ and lies in one of the three pants. By (Q) it is parallel to a, to c or to a puncture, which contradicts our assumption or the essentiality of γ.

Next we compute two intersection numbers of $f(a) = T_a^k T_d(a)$. Since $i(a, a) = i(a, c) = 0$, the twist T_a^k fixes a and c by (D). By (N) and (I1),

$$i(a, f(a)) = i(T_a^{-k}(a), T_d(a)) = i(a, T_d(a)) = i(d, a)^2 = 4.$$

By (N), $i(c, f(a)) = i(T_a^{-k}(c), T_d(a)) = i(c, T_d(a))$, and (I2) with $x = d$, $y = a$, $z = c$, $m = 1$ gives $\big|\, i(T_d(a), c) - i(d, a)\, i(d, c)\big| \le i(a, c) = 0$, so $i(c, f(a)) = 2 \cdot 2 = 4$.

Now let γ be any essential curve. If γ is disjoint from both a and c, it is a or c, and both meet $f(a)$. So some member of $\{a, c, f(a)\}$ meets γ, and this set, which lies in $A = \{f^m(a), f^m(c) : m \in \mathbb{Z}\}$, fills S.

(3) Two exceptional members, k apart. By (D), T_a and T_c commute, so $t^j = T_a^j T_c^{\varepsilon j}$ and

$$t^j f = T_a^{k+j}\, T_c^{\varepsilon j}\, T_d \qquad (j \in \mathbb{Z}).$$

For $j = 0$ this is $T_a^k T_d$. By Lemma 2, $i(d, C') = i(a, C') = 0$, so T_d and T_a fix C' by (D), and so does $T_a^k T_d$. For $j = -k$ it is $T_c^{-\varepsilon k} T_d$. By Lemma 2, $i(d, C) = i(c, C) = 0$, so T_d and T_c fix C, and so does $T_c^{-\varepsilon k} T_d$. Since C and C' are essential, neither member is pseudo-Anosov, by (P).

(4) Conclusion. For each $k \ge 1$ the hypothesis of Ivanov's question holds for (f_k, t), and the set of j with $t^j f_k$ not pseudo-Anosov contains 0 and $-k$. Any run of consecutive integers that contains both has at least $k + 1$ elements. If N_S existed, taking $k = N_S$ would give a contradiction. So no such N_S exists for the sphere with five boundary components. □

Remark 3 (conventions). With right-handed twists every T_x becomes T_x^{-1}. The proof uses only (D), (I1), (I2) and (N), which hold for both, so it applies verbatim. Since $f\, t^j = f\, (t^j f)\, f^{-1}$, the members $f t^j$ and $t^j f$ are conjugate and have the same exceptional j.

Remark 4 (what the example does not show). In the family $t^j f_k$ the exceptional set contains 0 and $-k$; we do not determine it, so the example bears only on where exceptions lie, not on how many there are. The base maps f_k are reducible, so the example does not touch the version of the question in which f is pseudo-Anosov. It also uses a filling hypothesis on the union of the orbits, as Ivanov states it; for the variant in which each orbit fills on its own it says nothing, since the f_k-orbit of a is disjoint from C' ($T_a^k T_d$ fixes C' and $i(a, C') = 0$).

5 What is checked by machine

The proof above is complete as written. Its finite combinatorial core is also checked by a Lean proof and by two independent programs, in the folders `lean/` and `checks/` that accompany this paper. We state what each covers.

The model. The fundamental group of S (the disk with four punctures) is free on loops $x_1, \dots, x_4$ around $p_1, \dots, p_4$, with $x_1x_2x_3x_4$ parallel to ∂D. The half twists act by the Artin automorphisms

$$\sigma_i : \; x_i \mapsto x_i x_{i+1} x_i^{-1}, \quad x_{i+1} \mapsto x_i, \quad x_m \mapsto x_m \; (m \neq i, i+1),$$

and the twists about the round curves are the full twists of the enclosed strands [4, Ch. 9]: $T_a = \sigma_1^2$, $T_d = \sigma_2^2$, $T_c = \sigma_3^2$, $T_{C'} = (\sigma_1\sigma_2)^3$, $T_C = (\sigma_2\sigma_3)^3$, up to one global choice of handedness. The curves are the conjugacy classes, up to inversion, of the words x_1x_2, x_2x_3, x_3x_4, $x_1x_2x_3$, $x_2x_3x_4$. In this model the theorem rests on five finite facts:

F0. Each of $T_a, T_d, T_c, T_{C'}, T_C$ fixes its own curve.

F1. The pentagon pattern of Lemma 2: $T_u(v) = v$ for the ten ordered disjoint pairs and $T_u(v) \neq v$ for the ten ordered meeting pairs.

F2. $T_a T_c = T_c T_a$.

F3. For every k and both signs, $T_a^k T_d$ fixes C' and $t^{-k} f_k = T_c^{-\varepsilon k} T_d$ fixes C.

F4. For every k, $T_a(f_k(a)) \neq f_k(a)$ and $T_c(f_k(a)) \neq f_k(a)$, that is, $f_k(a)$ meets a and c.

Lean. The file `lean/Multitwist.lean` (Lean 4 with Mathlib) proves F0–F4 for every $k \in \mathbb{N}$ and both signs, as statements about the Artin automorphisms of the free group F_4 and conjugacy in it. For disjoint pairs it gives explicit conjugators. For meeting pairs it gives a homomorphism $F_4 \to \mathrm{Sym}(3)$ that sends exactly one of $T_u(v)$ and v to the identity, so $T_u(v)$ is conjugate neither to v nor to v^{-1}. For every k, F3 follows by commuting and canceling twists (F2), and F4 by applying the automorphism T_a^k, which preserves conjugacy, to the case $k = 0$. The file `lean/Check.lean` restates each fact with its exact type, checks that exchanging C and C' makes the $j = 0$ statement false, and checks which axioms each statement uses: only the standard `propext`, `Classical.choice` and `Quot.sound`.

What Lean does *not* check is the surface topology, which is the written proof above: that the Artin automorphisms are the maps the twists induce on π_1; that essential curves correspond to conjugacy classes up to inversion, so that $T_u(v) = v$ in the model means that the curves are equal; that $T_u(v) = v$ if and only if $i(u, v) = 0$; the exact values $i = 2$ and $i = 4$; the pants argument for filling; and that a class fixing an essential curve is not pseudo-Anosov. Mathlib has no theory of mapping class groups or isotopy classes of curves, so these steps are not formalized.

Checker A. The program `checks/check_free_group.py` works in the same free-group model: it computes the Artin action on reduced words and compares curves by a canonical form of their cyclic words up to inversion. It verifies F0–F4 for $k = 1, \dots, 30$ and both signs, 327 checks in all. It is exact and uses only the Python standard library.

Checker B. The program `checks/check_dynnikov.py` was written separately, from the statement of the facts alone and without seeing checker A or the Lean file. It uses a different model: curves in the four-punctured disk are encoded by their Dynnikov coordinates [3] in $\mathbb{Z}^4$, a bijective encoding of multicurves, and the half twists act by the piecewise-linear update rules of Hall and Yurttaş [7]. Intersection numbers with round curves are computed exactly by the formula of Yurttaş and Hall [12, Lemma 9] and, independently, by the twist test $i(u, v) = 0 \iff T_u(v) = v$. Before checking F0–F4 it validates the model on known facts: inverses, the braid relations, triviality of the full twist, and the identity (I1) on random curves. It verifies F0–F4 for $k = 1, \dots, 30$ and both

signs, also with every twist reversed, and confirms the exact values $i(a, f_k(a)) = i(c, f_k(a)) = 4$. It is exact (integers only) and uses only the Python standard library.

Corruptions. Both checkers must reject four corrupted inputs: C replaced by $T_a(C)$; C' replaced by $T_c(C')$; C and C' exchanged; and c replaced by $T_{C'}^{-1}(c)$. Each corruption breaks the pentagon pattern and the fixed-curve facts, and both checkers reject all four. The command `python3 checks/check.py` runs both checkers and all corruptions in about a second.

The two programs implement one list of facts in two models, so they guard against errors of computation, not against an error in the topology above; like the Lean proof, they check the combinatorial core, while the step from it to the theorem is the written proof.

Acknowledgments

The preliminary checks used Mark Bell's programs `flipper` and `curver`. Before the checkers of Section 5 were written, `flipper` was used to check the two exceptional members for $k \leq 5$ and both signs, on two different triangulations, and `curver` to cross-check the curve facts F0–F4 computed by checker B. These checks are not part of the argument: neither the proof nor the final checkers, which use only the Python standard library, depend on them.

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